# Voltage dividers & sensors

ENGINEERING WITH MR KELLY

Higher · Electronics & control · Worksheet

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ calculate the voltages in a voltage divider
- ☐ rearrange the voltage divider formula to find a resistor value
- ☐ describe how a sensing circuit works using LDR or thermistor resistance and Vout
- ☐ design a voltage divider to switch at a given light level or temperature

## I DO — Worked example

Read how a full-mark answer is built

*A frost alarm has a thermistor at the bottom of a voltage divider. Its resistance rises as the temperature falls.*

**1 Describe how the circuit switches the alarm on when it gets cold.** (3 marks)

**Full-mark answer** each ✓ is one mark

As the temperature falls, the thermistor's resistance increases ✓. This means the voltage across the thermistor, Vout, increases ✓. When Vout reaches about 0.7 V the transistor switches on and current flows through the alarm ✓.

**Why marks are not awarded**

*"When it gets cold the thermistor detects it and turns on the alarm."*

0 marks: no reference to resistance or Vout.

## WE DO — Guided practice

Use the starters and word bank to help you

*A voltage divider has R1 = 10 kΩ and R2 = 15 kΩ across a 6 V supply.*

**2 Calculate V2.** (2 marks)

**Sentence starters**

One … is …, which means …

Another … is …, so …

**Word bank**

Tip: 2 marks = 2 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** A thermistor is at the bottom of a divider. As temperature rises, Vout...
☐ Rises ☐ Falls ☐ Stays the same ☐ Reverses

**b)** Vs = 9 V, R1 = 1 kΩ, R2 = 2 kΩ. V2 is...
☐ 3 V ☐ 4.5 V ☐ 6 V ☐ 9 V

**c)** A good description of a sensing circuit starts with...
☐ The relay ☐ The sensor's resistance changing ☐ The LED ☐ The battery

**d)** In a divider, the larger resistor has...
☐ The smaller share of voltage ☐ The larger share of voltage ☐ No voltage ☐ The same voltage

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Voltage divider rule** | ____ | **A** V1/V2 = R1/R2 |
| 2. **V out of a divider** | ____ | **B** Sensor resistance → Vin → transistor → relay → output. |
| 3. **Sensor at the bottom of a divider** | ____ | **C** RA/RB = R1/R2; no voltage difference across the middle. |
| 4. **LDR at the bottom** | ____ | **D** Vout rises as it gets darker (resistance rises). |
| 5. **Describing a sensing circuit** | ____ | **E** Vout rises when the sensor's resistance rises. |
| 6. **Balanced bridge** | ____ | **F** V2 = Vs × R2 ÷ (R1 + R2) |

*A thermistor is R2 in a voltage divider with a fixed resistor R1 = 4.7 kΩ and a 5 V supply. At 20 °C the thermistor is 2.2 kΩ.*

### 5 Calculate Vout across the thermistor at 20 °C. (2 marks)

*A light-sensing circuit needs V2 = 1.5 V from a 6 V supply. R2 (an LDR) is 1.2 kΩ at the switching light level.*

**6 Calculate the value of R1.** (3 marks)

*A Wheatstone bridge has two voltage dividers across 12 V: R1 = 10 kΩ, R2 = 10 kΩ; R3 = 8 kΩ, R4 = 12 kΩ.*

**7 Calculate the voltage between the two midpoints.** (3 marks)

**Check and reflect**

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?